

EOD – Jenkins-01 Setup and Configuration

Date: 28 September 2026

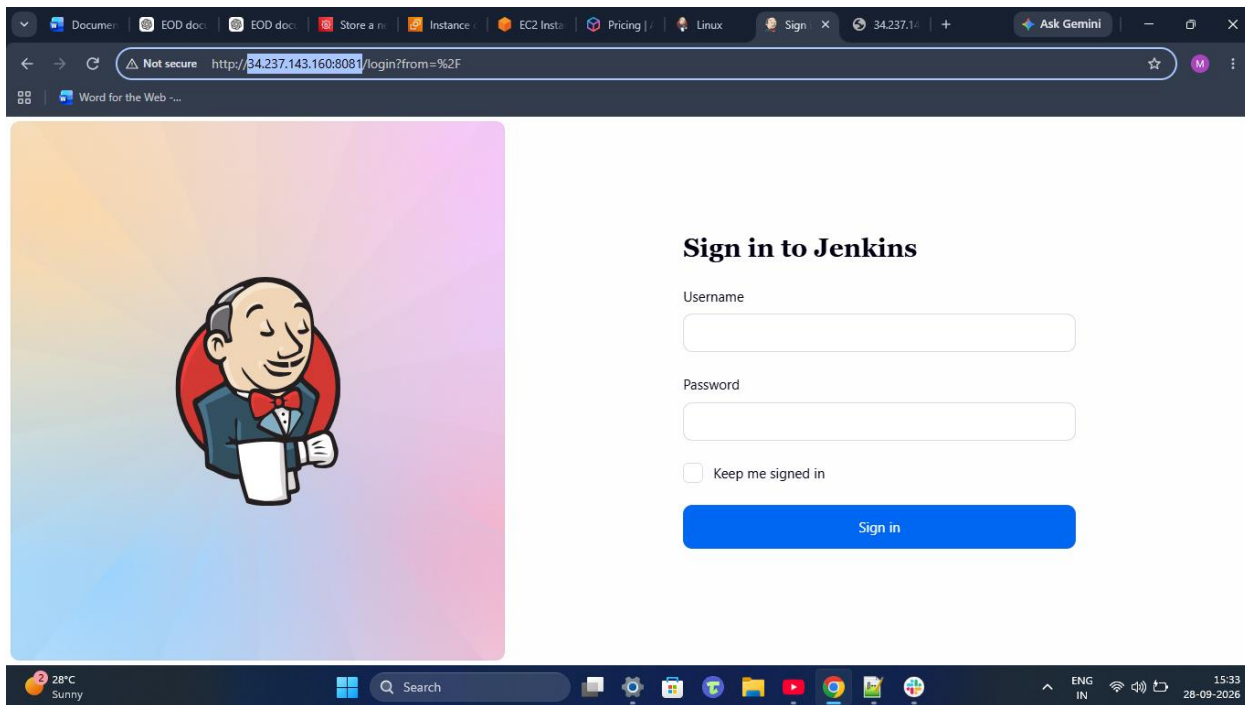
Task: Jenkins-01

Work Completed

Today I worked on the **Jenkins-01** setup and configuration activities.

1. Jenkins Installation

- Installed Jenkins on the Jenkins server.
- Configured Jenkins to run on **port 8081**.
- Verified that Jenkins was accessible through the configured port.



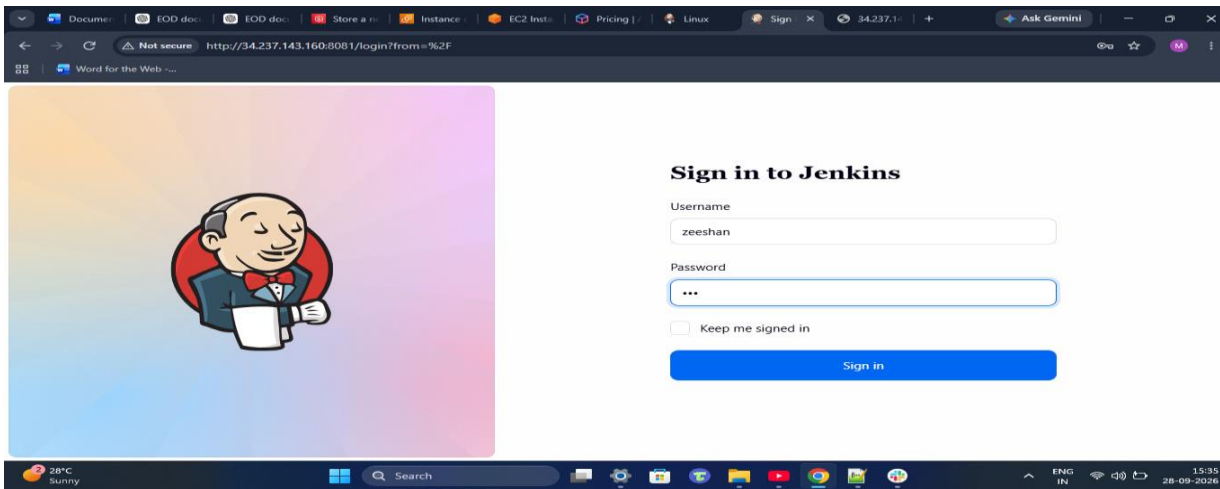
```
sudo vi /usr/lib/systemd/system/jenkins.service
```

```
sudo systemctl daemon-reload
```

```
sudo systemctl restart jenkins
```

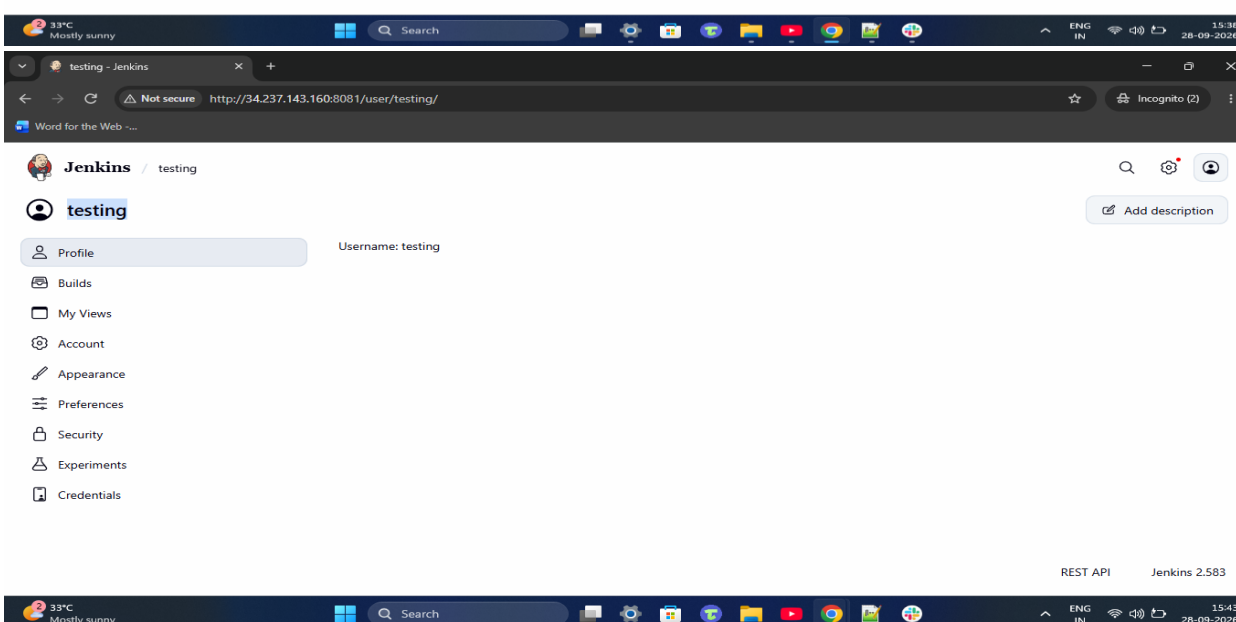
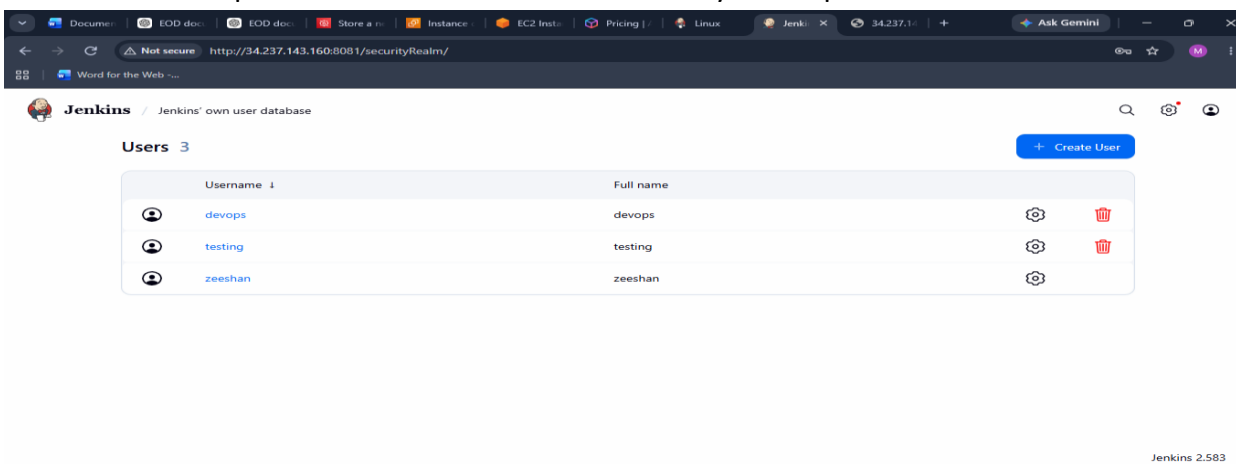
2. Jenkins Server Security

- Performed the basic security configuration for the Jenkins server.
- Enabled authentication and secured access to the Jenkins dashboard.
- Reviewed user access and permissions.



3. Jenkins Users and Access

- Created the following Jenkins users:
 - DevOps
 - Testing
- Configured limited access based on user requirements.
- Reviewed permissions to ensure users have only the required access.



4. Jenkins Labels and Job Restrictions

- Configured **labels** for Jenkins nodes/agents.
- Understood how labels are used to identify specific nodes.
- Configured jobs to execute only on nodes matching the required label.
- Tested the **“Restrict where this project can be run”** configuration.

The screenshot shows the Jenkins web interface for the 'Built-In Node'. The left sidebar contains links for Status, Configure, Build History, Load Statistics, and Script Console. The main content area shows the 'Built-In Node' status as 'Online'. A 'Monitoring Data' dropdown is visible. Below this, the 'Labels' section shows 'master'. The 'Projects tied to Built-In Node' section is currently empty. The bottom of the page shows the REST API and Jenkins version 2.583.

5. Sample Jenkins Jobs

Created sample Jenkins jobs using the provided GitHub repository:

Source Repository:

betawins/Techie_horizon_Login_app

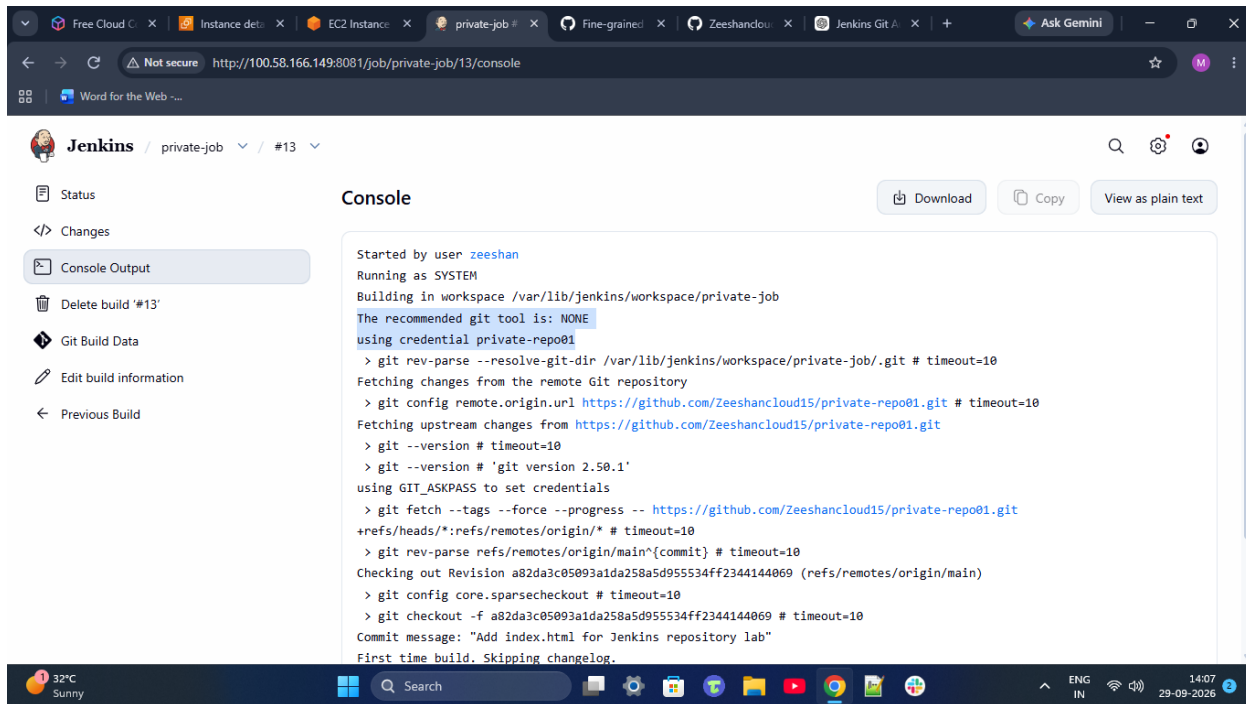
- Configured the repository as the source code.
- Configured Jenkins to fetch the source code from GitHub.
- Created and tested **three sample Jenkins jobs**.
- Verified job execution and source-code checkout.

The screenshot shows the Jenkins 'Build Queue' page. The left sidebar includes links for 'New Item', 'Build History', 'Build Queue (1)', and 'Build Executor Status'. The main content area displays a table of jobs in the queue. The table has columns for status, icon, name, last success, last failure, and last duration. Three jobs are listed: 'firstjob', 'secondjob', and 'third-job'. The 'firstjob' is completed, while 'secondjob' and 'third-job' are in the queue.

S	W	Name	Last Success	Last Failure	Last Duration
✓	☁	firstjob	46 min #2	47 min #1	3.8 sec
...	☀	secondjob	N/A	N/A	N/A
✓	☀	third-job	16 sec #1	N/A	10 ms

6. Jenkins Job with Private GitHub Repository

- Created an additional Jenkins job using a **private GitHub repository**.
- Configured GitHub authentication/credentials in Jenkins.
- Connected the Jenkins job to the private repository.
- Tested source-code checkout and job execution.



The screenshot shows a web browser window displaying the Jenkins console for a job named 'private-job' (build #13). The browser's address bar shows the URL 'http://100.58.166.149:8081/job/private-job/13/console'. The Jenkins interface includes a left sidebar with navigation options like 'Status', 'Changes', 'Console Output', 'Delete build', 'Git Build Data', 'Edit build information', and 'Previous Build'. The main area is titled 'Console' and contains the following output:

```
Started by user zeeshan
Running as SYSTEM
Building in workspace /var/lib/jenkins/workspace/private-job
The recommended git tool is: NONE
using credential private-repo01
> git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/private-job/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/Zeeshancloud15/private-repo01.git # timeout=10
Fetching upstream changes from https://github.com/Zeeshancloud15/private-repo01.git
> git --version # timeout=10
> git --version # 'git version 2.50.1'
using GIT_ASKPASS to set credentials
> git fetch --tags --force --progress -- https://github.com/Zeeshancloud15/private-repo01.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision a82da3c05093a1da258a5d955534ff2344144069 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f a82da3c05093a1da258a5d955534ff2344144069 # timeout=10
Commit message: "Add index.html for Jenkins repository lab"
First time build. Skipping changelog.
```

The bottom of the image shows a Windows taskbar with the date and time '14:07 29-09-2026' and system icons for network, volume, and language.